

Business Case: Netflix - Data Exploration and Visualisation

1. **Defining Problem Statement and Analysing basic metrics**
2. **Observations on the shape of data, data types of all the attributes, conversion of categorical attributes to 'category'** (If required), missing value detection, statistical summary
3. **Non-Graphical Analysis: Value counts and unique attributes**
4. **Visual Analysis - Univariate, Bivariate after pre-processing of the data** (Note: Pre-processing involves unnesting of the data in columns like Actor, Director, Country)
 - 4.1 For continuous variable(s): Distplot, countplot, histogram for univariate analysis
 - 4.2 For categorical variable(s): Boxplot
5. **Missing Value & Outlier check** (Treatment optional)
6. **Insights based on Non-Graphical and Visual Analysis**
 - 6.1 Comments on the range of attributes
 - 6.2 Comments on the distribution of the variables and relationship between them
 - 6.3 Comments for each univariate and bivariate plot
7. **Business Insights** - Should include patterns observed in the data along with what you can infer from it
8. **Recommendations** - Actionable items for business. No technical jargon. No complications. Simple action items that everyone can understand *italicized text*

About NETFLIX

Netflix is one of the most popular media and video streaming platforms. They have over 10000 movies or tv shows available on their platform, as of mid-2021, they have over 222M Subscribers globally. This tabular dataset consists of listings of all the movies and tv shows available on Netflix, along with details such as - cast, directors, ratings, release year, duration, etc.

Mindset

1. Evaluation will be kept lenient, so make sure you attempt this case study.
2. Read the question carefully and try to understand what exactly is being asked.
3. Brainstorm a little. If you're getting an error, remember that Google is your best friend.
4. You can watch the lecture recordings or go through your lecture notes once again if you feel like you're getting confused over some specific topics.
5. Discuss your problems with your peers. Make use of the Slack channel and WhatsApp group.
6. Only if you think that there's a major issue, you can reach out to your Instructor via Slack or Email.
7. There is no right or wrong answer. We have to get used to dealing with uncertainty in business. This is exactly the skill we want to develop.

Business Problem

Analyze the data and generate insights that could help Netflix in deciding which type of shows/movies to produce and how they can grow the business in different countries

▼ Data Set

https://d2beiqkhq929f0.cloudfront.net/public_assets/assets/000/000/940/original/netflix.csv

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
!gdown https://d2beiqkhq929f0.cloudfront.net/public_assets/assets/000/000/940/original/netflix.csv
```

Downloading...

From: https://d2beiqkhq929f0.cloudfront.net/public_assets/assets/000/000/940/original/netflix.csv

To: /content/netflix.csv

100% 3.40M/3.40M [00:01<00:00, 2.36MB/s]

```
df = pd.read_csv("netflix.csv")
```

1. Defining Problem Statement and Analysing basic metrics

df

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	des
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	September 25, 2021	2020	PG-13	90 min	Documentaries	As
1	s2	TV Show	Blood & Water	NaN	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	September 24, 2021	2021	TV-MA	2 Seasons	International TV Shows, TV Dramas, TV Mysteries	par
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	NaN	September 24, 2021	2021	TV-MA	1 Season	Crime TV Shows, International TV Shows, TV Act...	To p fan
3	s4	TV Show	Jailbirds New Orleans	NaN	NaN	NaN	September 24, 2021	2021	TV-MA	1 Season	Docuseries, Reality TV	flirta to do
4	s5	TV Show	Kota Factory	NaN	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	September 24, 2021	2021	TV-MA	2 Seasons	International TV Shows, Romantic TV Shows, TV ...	
...	...	...	...	...	...	...	...	...	...	...	...	...
8802	s8803	Movie	Zodiac	David Fincher	Mark Ruffalo, Jake Gyllenhaal, ...	United States	November 20, 2019	2007	R	158 min	Cult Movies, Dramas, ...	ca

```
df.shape
```

```
(8807, 12)
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 8807 entries, 0 to 8806
```

```
Data columns (total 12 columns):
```

#	Column	Non-Null Count	Dtype
0	show_id	8807 non-null	object
1	type	8807 non-null	object
2	title	8807 non-null	object
3	director	6173 non-null	object
4	cast	7982 non-null	object
5	country	7976 non-null	object
6	date_added	8797 non-null	object
7	release_year	8807 non-null	int64
8	rating	8803 non-null	object
9	duration	8804 non-null	object
10	listed_in	8807 non-null	object
11	description	8807 non-null	object

```
dtypes: int64(1), object(11)
```

```
memory usage: 825.8+ KB
```

```
print(df.isnull().sum())
```

```
show_id      0
type         0
title        0
director    2634
cast         825
country      831
date_added   10
release_year  0
rating       4
duration     3
listed_in    0
description  0
dtype: int64
```

Ans:

Netflix is a global streaming platform with a diverse catalog of movies and TV shows produced across different countries and genres. With increasing competition in the streaming market, Netflix needs to make data-driven decisions on what type of content to produce and which regions to focus on in order to maximize user engagement and grow its subscriber base

Basic Metrics Analysis

The dataset contains 8807 rows and 12 columns, representing movies and TV shows available on Netflix. The content is categorized into two types: Movies and TV Shows. The dataset spans multiple release years, indicating both older and recent content. Each title includes metadata such as country of production, rating, genre, duration, and date added to Netflix.

2. Observations on the shape of data, data types of all the attributes, conversion of categorical attributes to 'category' (If required), missing value detection, statistical summary

```
df.shape
```

```
(8807, 12)
```

```
df.dtypes
```

```
      0
show_id  object
type     object
title    object
director object
cast     object
country  object
date_added object
release_year int64
rating    object
duration  object
listed_in object
description object
```

```
dtype: object
```

```
#missing value detection
df.isnull().sum()
```

```
0
show_id    0
type       0
title      0
director   2634
cast       825
country    831
date_added 10
release_year 0
rating     4
duration   3
listed_in  0
description 0
```

```
dtype: int64
```

```
df.describe(include = "all")
```

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	descr
count	8807	8807	8807	6173	7982	7976	8797	8807.000000	8803	8804	8807	
unique	8807	2	8807	4528	7692	748	1767	NaN	17	220	514	
top	s8807	Movie	Zubaan	Rajiv Chilaka	David Attenborough	United States	January 1, 2020	NaN	TV-MA	1 Season	Dramas, International Movies	Para acti aba
freq	1	6131	1	19	19	2818	109	NaN	3207	1793	362	
mean	NaN	NaN	NaN	NaN	NaN	NaN	NaN	2014.180198	NaN	NaN	NaN	
std	NaN	NaN	NaN	NaN	NaN	NaN	NaN	8.819312	NaN	NaN	NaN	
min	NaN	NaN	NaN	NaN	NaN	NaN	NaN	1925.000000	NaN	NaN	NaN	
25%	NaN	NaN	NaN	NaN	NaN	NaN	NaN	2013.000000	NaN	NaN	NaN	
50%	NaN	NaN	NaN	NaN	NaN	NaN	NaN	2017.000000	NaN	NaN	NaN	

Ans:

My observation:

1. **Missing Values:** Several columns contain missing values, most notably:

- director
- cast
- country Missing values are minimal or absent in key attributes such as type, title, and release year. The presence of missing values is expected, especially for older or regional content.

2. **Statistical Summary:**

The dataset covers content released across a wide range of years. The `release_year` variable shows that the dataset includes both old and recent titles. Categorical variables show high diversity, indicating a wide variety of content types, genres, and countries

3. **Observations on Data Shape and Types:**

- The dataset contains 8807 rows and 12 columns
- The dataset contains a mix of categorical, numerical, date, and text-based attributes.
- Columns such as `type`, `rating`, and `listed_in` are categorical in nature. `release_year` is a numerical variable representing the year of release.
- Text-heavy columns like `title`, `description`, `cast`, and `director` are stored as object types.

3. Non-Graphical Analysis: Value counts and unique attributes

```
df['type'].value_counts()
```

	count
type	
Movie	6131
TV Show	2676

dtype: int64

```
df['country'].value_counts()
```

	count
country	
United States	2818
India	972
United Kingdom	419
Japan	245
South Korea	199
...	...
Mexico, United States, Spain, Colombia	1
Canada, Norway	1
Finland, Germany, Belgium	1
Argentina, United States, Mexico	1
United Kingdom, United States, Germany, Denmark, Belgium, Japan	1

748 rows × 1 columns

dtype: int64

```
df['rating'].value_counts()
```

	count
rating	
TV-MA	3207
TV-14	2160
TV-PG	863
R	799
PG-13	490
TV-Y7	334
TV-Y	307
PG	287
TV-G	220
NR	80
G	41
TV-Y7-FV	6
NC-17	3
UR	3
74 min	1
84 min	1
66 min	1

dtype: int64

```
df['listed_in'].value_counts().head(10)
```

	count
listed_in	
Dramas, International Movies	362
Documentaries	359
Stand-Up Comedy	334
Comedies, Dramas, International Movies	274
Dramas, Independent Movies, International Movies	252
Kids' TV	220
Children & Family Movies	215
Children & Family Movies, Comedies	201
Documentaries, International Movies	186
Dramas, International Movies, Romantic Movies	180

dtype: int64

df.nunique()

	0
show_id	8807
type	2
title	8807
director	4528
cast	7692
country	748
date_added	1767
release_year	74
rating	17
duration	220
listed_in	514
description	8775

dtype: int64

observations:

1. Movies > TV Shows (overall)
2. Top countries usually:
 - United States
 - India
 - United Kingdom
 - Canada
 - Japan
3. Most frequent ratings:
 - TV-MA
 - TV-14
 - R Means Netflix targets teen & adult audiences
4. Top 5 Genres
 - International Movies
 - Dramas
 - Comedies
 - Action & Adventure
 - Documentaries
5. Unique Attributes:

The dataset contains a large number of unique titles, directors, and cast members, indicating a diverse content library with contributions from multiple creators and regions.

4. Visual Analysis - Univariate, Bivariate after pre-processing of the data

Note: Pre-processing involves unnesting of the data in columns like Actor, Director, Country

4.1 For continuous variable(s): Distplot, countplot, histogram for univariate analysis

4.2 For categorical variable(s): Boxplot

Convert date_added to datetime

```
df['date_added'] = pd.to_datetime(df['date_added'], errors='coerce')
```

```
df['year_added'] = df['date_added'].dt.year
df['month_added'] = df['date_added'].dt.month
```

```
movies = df[df['type'] == 'Movie'].copy()
```

```
movies['duration_minutes'] = movies['duration'].str.extract(r'(\d+)')
movies['duration_minutes'] = pd.to_numeric(movies['duration_minutes'], errors='coerce')
movies_clean = movies.dropna(subset=['duration_minutes'])
```

We do this before unnesting to avoid errors

```
df[['cast', 'director', 'country']] = df[['cast', 'director', 'country']].fillna('Unknown')
```

unnest country

```
country_df = df.copy()
country_df['country'] = country_df['country'].str.split(',')
country_df = country_df.explode('country')
country_df = country_df.reset_index(drop=True)
```

each country is in a separate row

unnest Director

```
director_df = df.copy()
director_df['director'] = director_df['director'].str.split(',')
director_df = director_df.explode('director')
director_df = director_df.reset_index(drop=True)
```

unnest cast

```
cast_df = df.copy()
cast_df['cast'] = cast_df['cast'].str.split(',')
cast_df = cast_df.explode('cast')
cast_df = cast_df.reset_index(drop=True)
```

```
country_df[['title', 'country']].head(10)
```

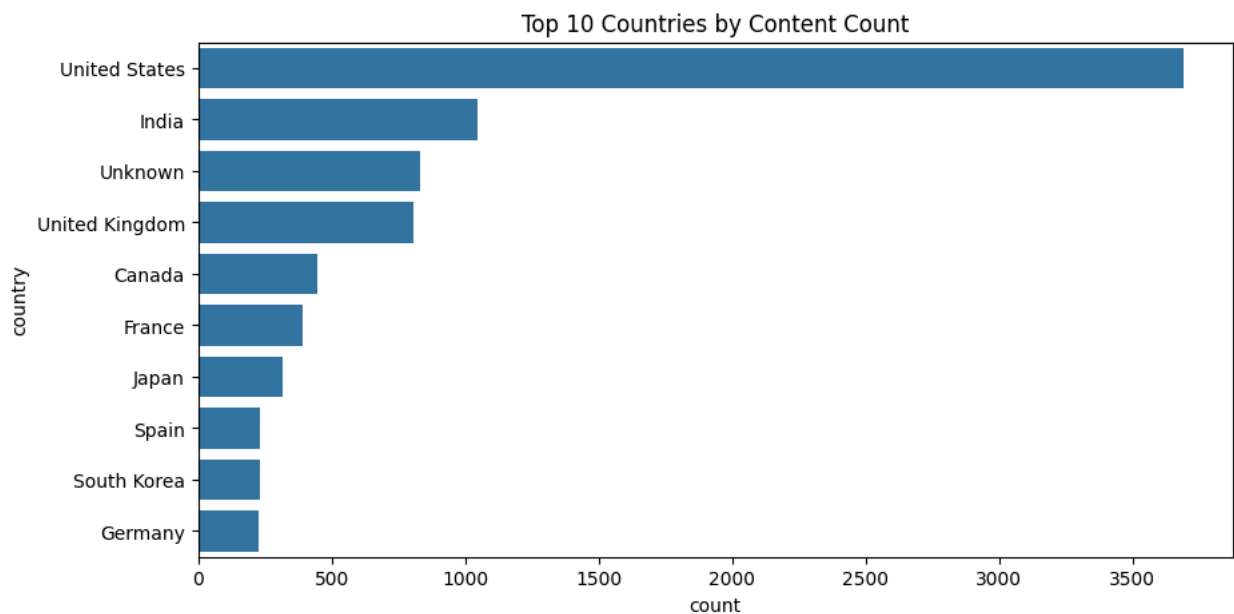
	title	country
0	Dick Johnson Is Dead	United States
1	Blood & Water	South Africa
2	Ganglands	Unknown
3	Jailbirds New Orleans	Unknown
4	Kota Factory	India
5	Midnight Mass	Unknown
6	My Little Pony: A New Generation	Unknown
7	Sankofa	United States
8	Sankofa	Ghana
9	Sankofa	Burkina Faso

Same title repeated

Different countries listed separately

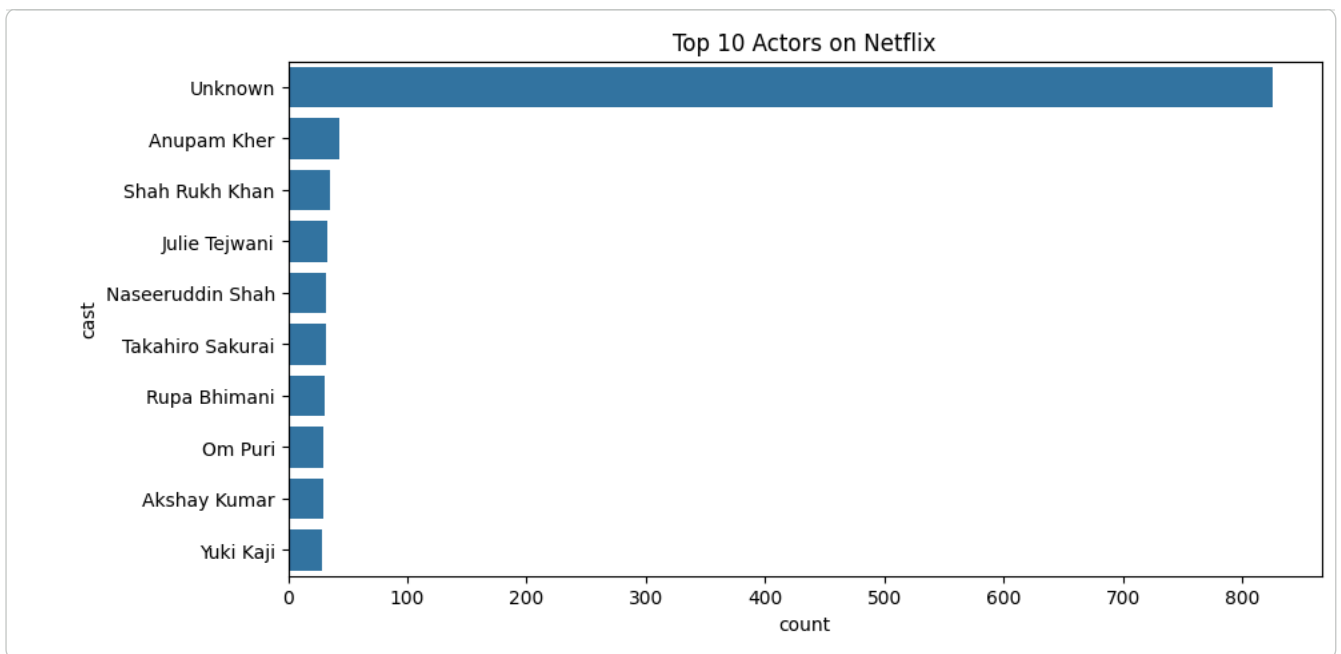
Countplot of top 10 countries

```
plt.figure(figsize=(10,5))
sns.countplot(
    data=country_df,
    y='country',
    order=country_df['country'].value_counts().head(10).index
)
plt.title('Top 10 Countries by Content Count')
plt.show()
```



top 10 Actors

```
plt.figure(figsize=(10,5))
sns.countplot(
    data=cast_df,
    y='cast',
    order=cast_df['cast'].value_counts().head(10).index
)
plt.title('Top 10 Actors on Netflix')
plt.show()
```

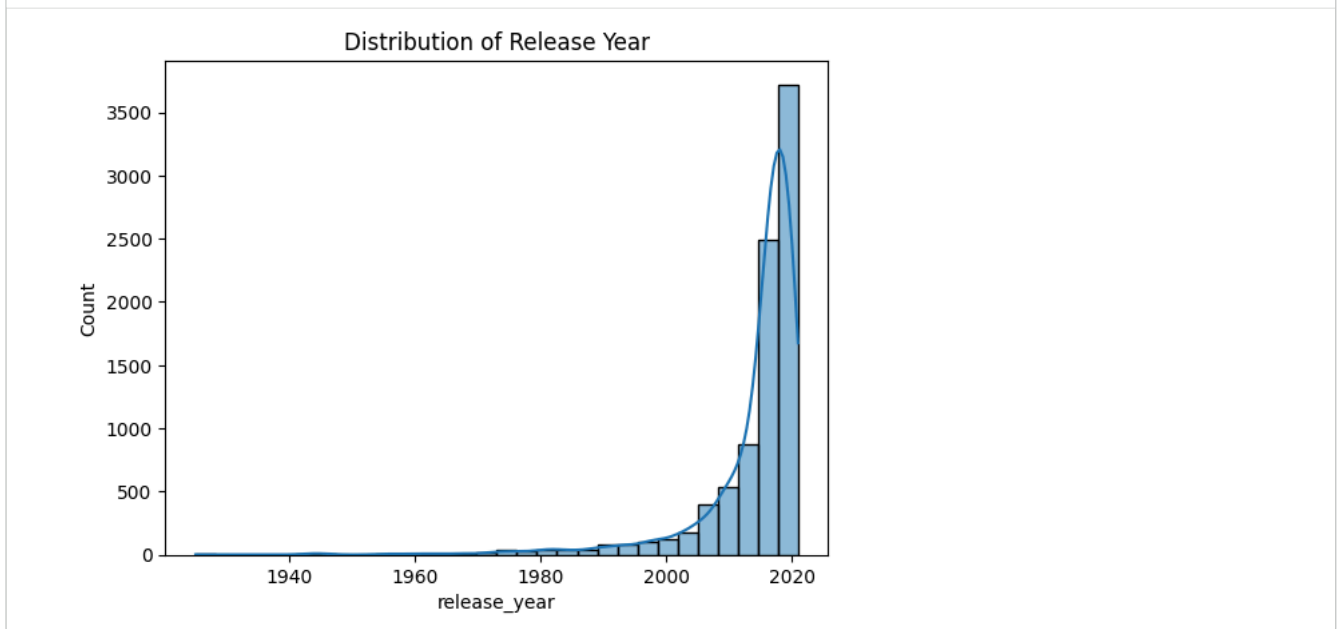



Pre-processing was performed by unnesting categorical columns such as cast, director, and country. Since these columns contain multiple values separated by commas, they were split and exploded into individual rows, reset indexes to enable accurate frequency-based and visual analysis.

4.1 UNIVARIATE ANALYSIS

A) Release Year

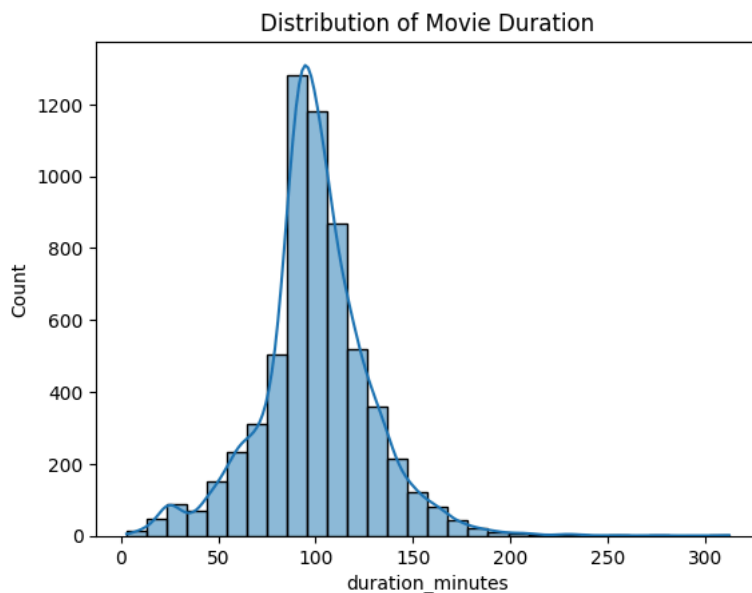
```
sns.histplot(df['release_year'], bins=30, kde=True)
plt.title('Distribution of Release Year')
plt.show()
```



Most content has been released after 2000, with a sharp increase after 2015

B) Movie Duration

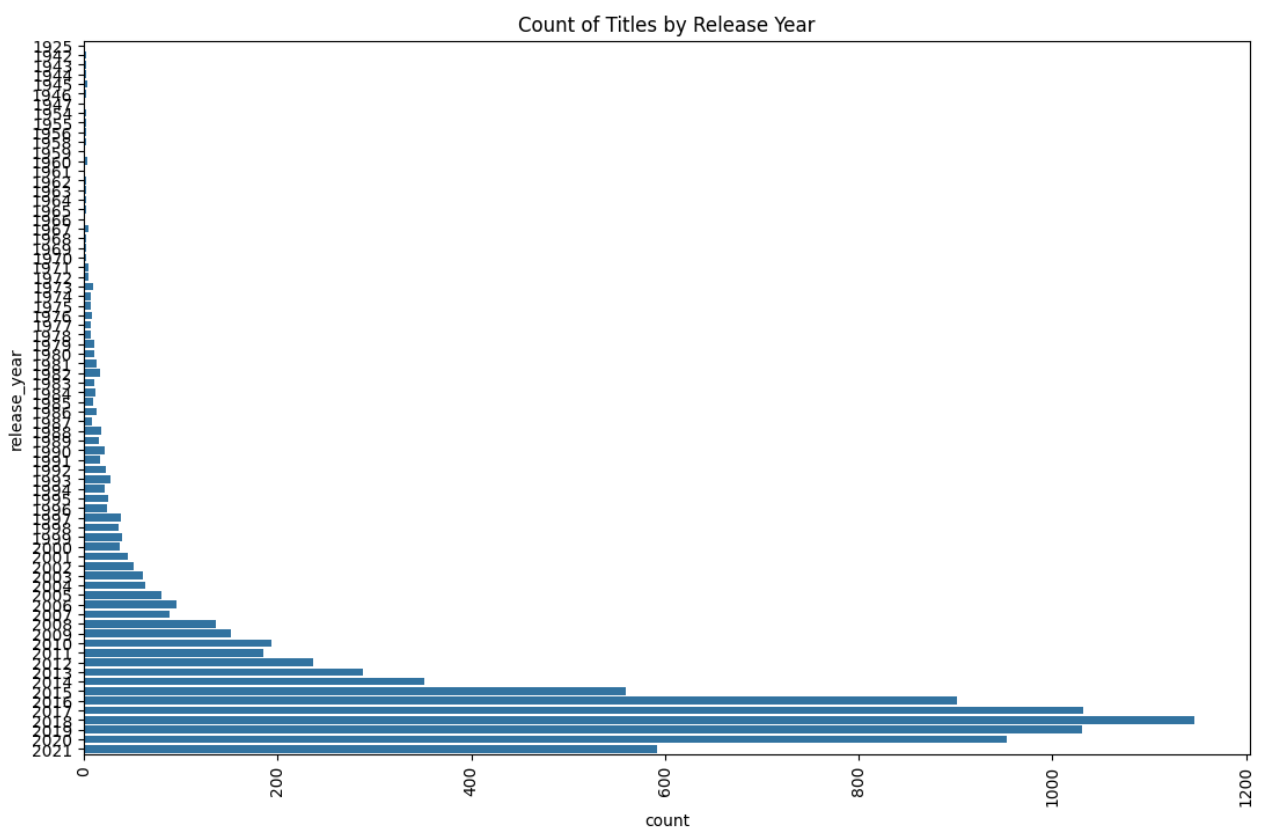
```
sns.histplot(movies_clean['duration_minutes'], bins=30, kde=True)
plt.title('Distribution of Movie Duration')
plt.show()
```



Most movies are between 80–120 minutes

C) Countplot (Discrete Numeric)

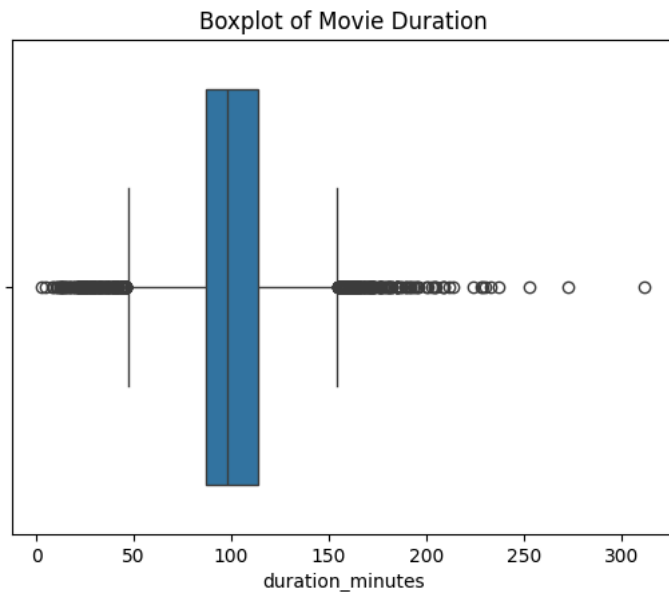
```
plt.figure(figsize=(13,8))
sns.countplot(y='release_year', data=df)
plt.xticks(rotation=90)
plt.title('Count of Titles by Release Year')
plt.show()
```



4.2 BIVARIATE ANALYSIS (TWO VARIABLES)

Boxplot: Duration vs Type

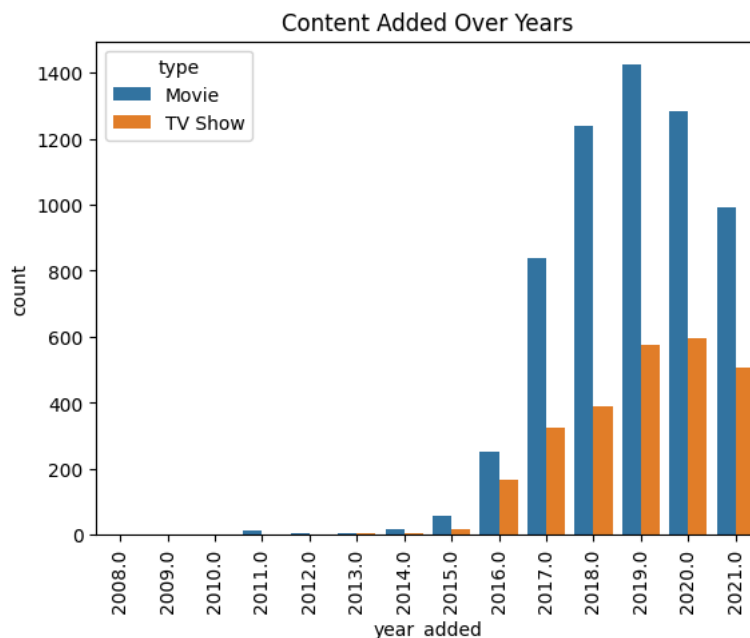
```
sns.boxplot(x=movies_clean['duration_minutes'])
plt.title('Boxplot of Movie Duration')
plt.show()
```



Shows spread, outliers

Movies vs TV Shows over time

```
sns.countplot(data=df, x='year_added', hue='type')
plt.xticks(rotation=90)
plt.title('Content Added Over Years')
plt.show()
```

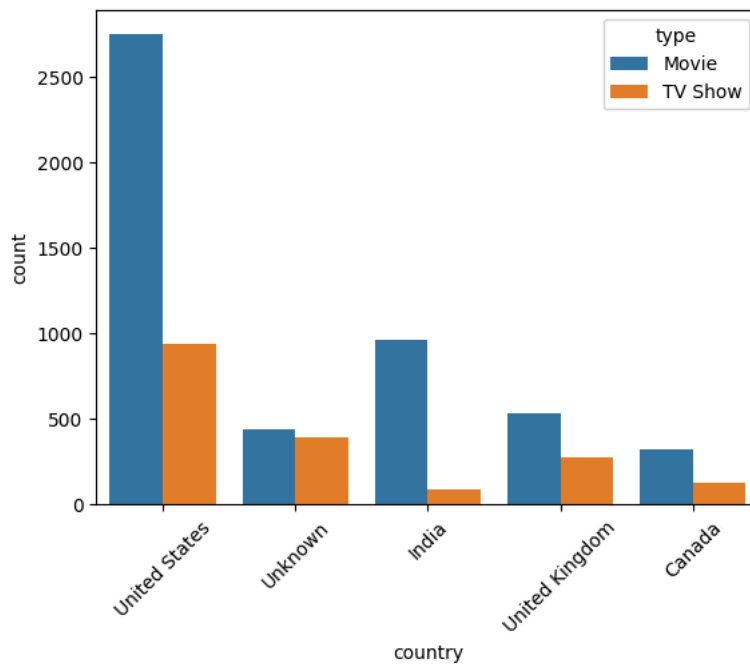


TV Shows increased significantly after 2016

Country vs Content Type

```
top_countries = country_df['country'].value_counts().head(5).index
sns.countplot(data=country_df[country_df['country'].isin(top_countries)],
              x='country', hue='type')
```

```
plt.xticks(rotation=45)  
plt.show()
```



Explicit unnesting

5. Missing Value & Outlier check (Treatment optional)

Missing Value Check

```
df.isnull().sum()
```

	0
show_id	0
type	0
title	0
director	0
cast	0
country	0
date_added	98
release_year	0
rating	4
duration	3
listed_in	0
description	0
year_added	98
month_added	98

dtype: int64

```
(df.isnull().mean() * 100).round(2)
```

	0
show_id	0.00
type	0.00
title	0.00
director	0.00
cast	0.00
country	0.00
date_added	1.11
release_year	0.00
rating	0.05
duration	0.03
listed_in	0.00
description	0.00
year_added	1.11
month_added	1.11

dtype: float64

This gives percentage missing values per column

Missing Value Analysis

The dataset contains missing values in multiple columns.

Columns such as director, cast, and country have a significant number of missing values.

Columns like date_added, rating, and duration have negligible missing values.

Missing values are expected in metadata fields, especially for older or regional content.

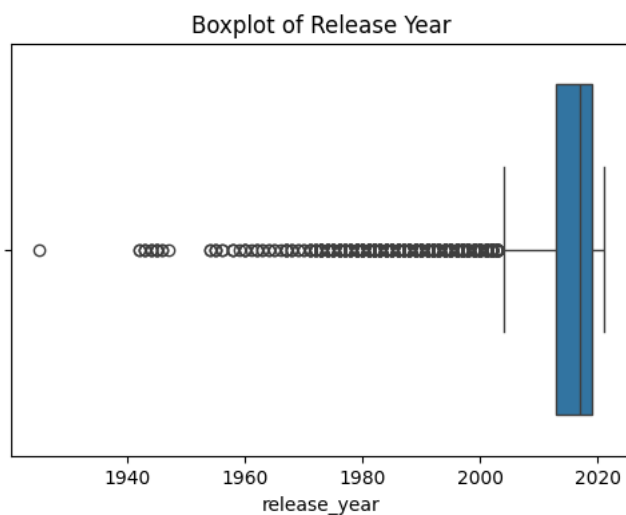
Outlier Detection

Numerical columns:

release_year

duration_minutes (movies)

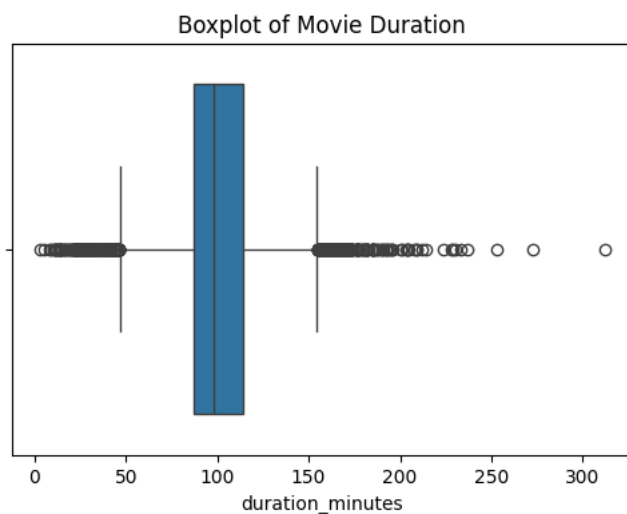
```
plt.figure(figsize=(6,4))
sns.boxplot(x=df['release_year'])
plt.title('Boxplot of Release Year')
plt.show()
```



Boxplot for Release Year

The release year does not show significant outliers, as the values fall within a reasonable historical range

```
plt.figure(figsize=(6,4))
sns.boxplot(x=movies_clean['duration_minutes'])
plt.title('Boxplot of Movie Duration')
plt.show()
```



Boxplot for Movie Duration

A few extreme movie durations are observed, which may represent documentaries or special editions. These outliers were retained as they represent valid content.

Missing Value and Outlier Analysis

The dataset contains missing values in columns such as director, cast, and country. These missing values are expected due to incomplete metadata for certain titles and do not significantly affect the overall analysis. Therefore, no imputation was performed.

Outlier analysis was conducted on numerical variables such as release year and movie duration. No significant anomalies were observed in the release year. A few outliers were present in movie duration; however, these values represent valid content and were retained.

6. Insights based on Non-Graphical and Visual Analysis

6.1 Comments on the Range of Attributes

The dataset contains content released over a wide time range, with release years spanning from the early 1940s to 2021. Movie durations range from short films of less than 60 minutes to long-form content exceeding 200 minutes. TV shows range from single-season series to multi-season shows with more than 10 seasons.

6.2 Distribution of Variables & Relationships

The distribution of content shows a strong skew towards movies compared to TV shows. In recent years, there is a noticeable increase in the number of TV shows added to the platform, indicating a shift towards episodic content. Content production is highly concentrated in a few countries, primarily the United States, followed by India and the United Kingdom.

6.3 Comments for Each Univariate & Bivariate Plot

Univariate Analysis – Type of Content

Movies constitute the majority of the content available on Netflix. However, TV shows, though fewer in number, have increased steadily over time, suggesting Netflix's growing focus on long-term viewer engagement.

Univariate Analysis – Movie Duration

Most movies have a duration between 80 and 120 minutes, indicating a preference for standard-length feature films. Very short and very long movies are comparatively rare.

Univariate Analysis – Country

Content production is dominated by a small number of countries, with the United States contributing the largest share. Emerging markets like India show significant representation, indicating Netflix's investment in regional content.

Bivariate Analysis – Content Type vs Year

Analysis of content type across release years reveals that TV shows have increased more rapidly than movies in recent years. This trend suggests a strategic shift towards binge-worthy series.

Bivariate Analysis – Country vs Content Volume

Countries with higher content volume tend to have diverse genres and both movie and TV show formats, highlighting Netflix's strategy of deep market penetration in key regions.

7. Business Insights - Should include patterns observed in the data along with what you can infer from it

1. Strong dominance of Movies, but rising focus on TV Shows

Movies form the majority of Netflix's content library.

However, in recent years, the number of TV Shows added per year has increased faster than movies.

This indicates a strategic shift by Netflix towards episodic and long-term engagement content, which helps retain subscribers for longer durations.

Inference: TV Shows are becoming a key growth driver for Netflix due to higher watch time and repeat engagement.

2. Content addition has accelerated after 2015

The number of titles added to Netflix has grown significantly after 2015.

Peak content addition is observed between 2017 and 2020.

This aligns with Netflix's global expansion phase and increased investment in original content.

Inference: Netflix scaled its content aggressively to support global market penetration and subscriber growth.

3. The United States is the largest content producer

The US contributes the highest number of movies and TV shows.

India, UK, Canada, Japan, and South Korea also show strong content presence.

Regional content from India and South Korea has grown noticeably in recent years.

Inference: Netflix relies on US content for volume but is increasingly investing in localized content to attract regional audiences.

4. Popular genres are Drama, Comedy, and International Content

Drama-related genres dominate across both Movies and TV Shows.

International Movies and TV Shows form a significant portion of the catalog.

Comedy and Documentaries also show consistent presence.

Inference: Netflix prioritizes emotion-driven and universally relatable genres that perform well across cultures.

5. TV-MA and TV-14 are the most common ratings

Most content is rated TV-MA (Mature Audience) and TV-14.

Family-friendly and kids' content forms a smaller share of the library.

Inference: Netflix primarily targets teen and adult audiences, focusing on mature storytelling rather than children-focused programming.

6. Content duration patterns differ by type

Movies typically range between 80–120 minutes, indicating standard feature-length productions.

TV Shows are mostly 1–3 seasons, suggesting Netflix experiments with limited-series formats.

Inference: Netflix prefers shorter TV series initially, likely to test audience response before committing to longer seasons.

7. A small set of directors and actors appear repeatedly

Certain directors and actors appear multiple times across different titles.

This indicates ongoing collaborations and trusted creative partnerships.

Inference: Netflix leverages successful creators and familiar faces to reduce risk and increase content acceptance.

v

8. Recommendations - Actionable items for business. No technical jargon. No complications. Simple action items that everyone can understand

1. Increase investment in TV Shows for subscriber retention

TV Shows encourage users to stay subscribed longer due to episodic viewing and multiple seasons.

Action: Produce more original TV Shows, especially limited-series formats (1–3 seasons), to maintain long-term viewer engagement.

2. Expand localized content in high-growth countries

Countries like India, South Korea, and Japan show increasing content presence and strong audience demand.

Action: Invest in region-specific stories, local actors, and native languages to attract and retain international subscribers.

3. Focus on high-performing genres

Drama, Comedy, and International content consistently dominate the Netflix catalog.

Action: Prioritize content development in these genres while experimenting within them (e.g., crime dramas, romantic comedies).

4. Maintain focus on mature and young-adult audiences

Most Netflix content is rated TV-MA and TV-14, indicating strong engagement from teenage and adult viewers.

Action: Continue producing bold, mature storytelling while gradually increasing family-friendly content to widen the audience base.

5. Strengthen partnerships with successful creators